FEASIBILITY REPORT

Terrain-Aware Multimodal Spatiotemporal Graph Neural Network for Probabilistic Flood Early Warning in Sri Lanka

Report contents

The report is structured to separate what is already evidenced in the repository from what remains proposed or to be validated.

1. Executive decision and scope

1.1 Decision statement

Proceed, with a staged scope. The strongest near-term deliverable is a leakage-controlled Sri Lankan flood benchmark plus a transparent sequence of baseline, graph, calibration, and multimodal experiments. The project is planned to present as terrain-aware and hydrology-grounded, not physics-guided in the strict sense.

1.2 Feasibility by dimension

Dimension	Assessment	Rationale
Research question	High	Clear methodological gaps and a testable response: topology, leakage control, calibration, and multimodal fusion.
Dataset and reproducibility	High with controls	410,931 measured node-days, public no-key sources, resumable scripts, and pre-computed split artifacts. Requires documentation and version freeze.
Core hydro-meteorological model	Medium-high	Inputs, targets, graph, losses, baselines, and metrics are specified. Performance remains unknown until trained.

1.3 Intended outcome

The intended outcome is a reproducible, probabilistic, node-level early-warning benchmark for Sri Lanka that supports comparison of persistence, climatology, discharge rules, tree models, sequence models, graph models, calibration methods, and a constrained SAR case study.

This is a research and decision-support outcome. It is not yet a public safety service, a surveyed inundation map, or a replacement for national warning authorities.

2. Feasibility assessment

2.1 Why the project is feasible now

The project has crossed the most difficult foundation threshold: the data construction problem is sufficiently specified to reproduce a large, multi-basin panel with explicit labels, graph structure, and evaluation partitions. The current repository contains the design needed to move from data engineering into controlled modeling.

- The core panel spans 51 hydrologically anchored nodes across 16 basins and 2003-2025, with 2025 coverage explicitly limited to three Kelani nodes.
- The label engine is causal: future targets are computed within node boundaries, and the 30-day warm-up is enforced through valid_sample.
- Leakage controls are encoded as data artifacts rather than left to each experiment: temporal split, Gin-basin holdout, and event_id groups.
- The graph keeps directed flow edges separate from distance-decayed spatial edges, enabling relational ablations.
- All listed acquisition sources are public and require no API key; downloads are cached, resumable, and rate-limit-aware.

2.2 What remains uncertain

The feasibility case is strong for research execution but not for outcome claims. The project still needs evidence on predictive skill, probability calibration, event lead time, generalisation to the Gin holdout, and the incremental value of SAR relative to scalar hydrologic features.

A second uncertainty is label validity. The primary label is a per-node GloFAS discharge exceedance, not observed inundation. The validation table shows a useful but bounded relationship to documented floods: 8 of 11 events fire at the high threshold and 9 of 11 at the moderate threshold, with the misses concentrated in pre-2011 south-western flash floods.

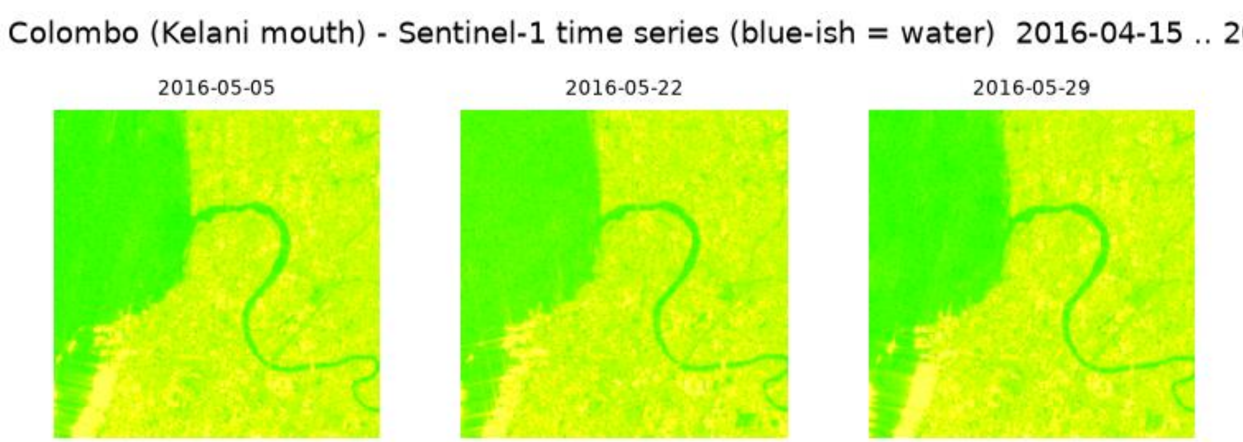
2.3 Feasibility boundary

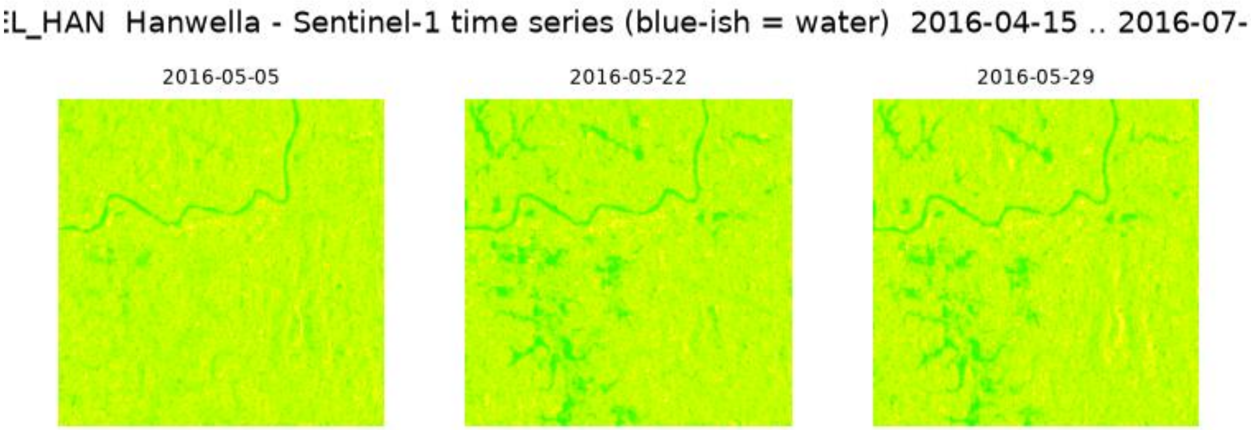
Boundary condition: The report supports a go decision for benchmark completion and model validation. It does not support a go decision for public operational deployment. That second decision requires a separate evidence package built around gauges, observed inundation, sub-daily forecasting, calibrated alert thresholds, and operational monitoring.

3. Evidence base: dataset, labels, and graph

3.1 Measured dataset position

Measure	Current value
Rows (node-days)	410,931
Columns	67
Valid supervised samples	409,350
Nodes / basins	51 / 16
Date span	2003-01-01 to 2025-12-31
Flood-state prevalence	1.907% on valid rows
Flood events	1,469 merged episodes
Event duration	Mean 5.39 days; median 3; IQR 2-6; maximum 45
Graph edges	239 total for 1 selected area





3.2 Split protocols

Protocol	Partition	N	Positive rate
Temporal	Train <= 2017-12-31	277,899	2.028%
Temporal	Validation 2018-2020	55,896	0.810%
Temporal	Test 2021 onward	75,555	2.275%
Basin holdout	Train: 15 basins	377,330	Not reported
Basin holdout	Test: Gin basin, 3 nodes	32,020	Not reported
Event groups	GroupKFold using event_id	1,469 groups	Not reported

Interpretation: The 2018-2020 validation block is genuinely quieter than train and test. Early stopping on validation PR-AUC can therefore be optimistic-biased; event-level cross-validation should be used as a model-selection safeguard.

4. Technical design and implementation feasibility

4.1 Acquisition pipeline

The acquisition design is technically feasible because it is decomposed into resumable stages: raw downloads, feature engineering, label and split creation, graph construction, validation, and optional imagery. The pipeline does not depend on a single opaque data dump; its scripts and compact artifacts are intended to support a rebuild.

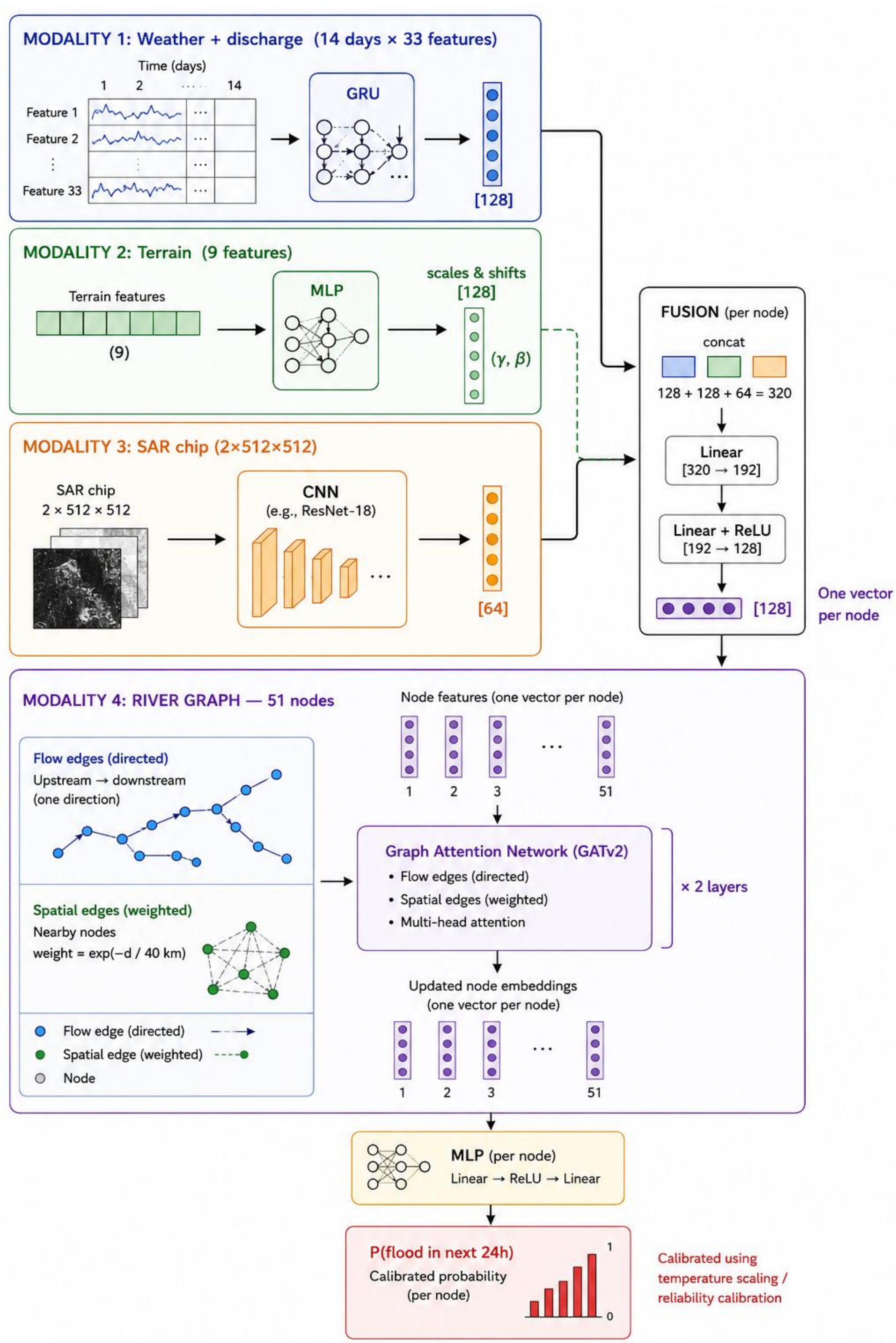
Modality	Source and role
Weather and precipitation	Open-Meteo Archive / ERA5-Land; NASA POWER fallback and comparison
Discharge	Open-Meteo Flood API / Copernicus GloFAS reanalysis

Modality	Source and role
Terrain	Copernicus DEM via Open-Meteo
SAR imagery	Microsoft Planetary Computer STAC / Sentinel-1 RTC
Topology	Hand-built node and downstream_of definitions

4.2 Proposed model architecture

The proposed system is a three-stream model with a fusion head. A temporal encoder processes a rolling k-day sequence of 33 dynamic features; a static MLP produces FiLM parameters to modulate the temporal representation; and an optional two-channel SAR CNN produces a pooled image embedding. The node representations are then passed through a relational graph layer with separate handling for flow and spatial edges.

Component	Design
Dynamic stream	GRU, TCN, or temporal attention over k=7, 14, or 30 days
Static stream	Elevation and log drainage proxy; optional zone and position one-hot features
SAR stream	Light CNN over 2 x 512 x 512 dB chips; presence mask for imageless nodes
Spatial reasoning	Two-layer relational GNN / edge-conditioned GraphSAGE or R-GAT
Outputs	t+1, t+2, t+3 flood probabilities; onset; next-day discharge; three-day maximum z-score
Loss	Focal BCE with label_confidence weighting plus Huber regression terms


MODALITY 1: Weather + discharge (14 days × 33 features)
Time (days)
1 2 ... 14
Feature 1
Feature 2
Feature 33
GRU
[128]
MODALITY 2: Terrain (9 features)
Terrain features
(9)
MLP
scales & shifts
[128]
(γ, β)
MODALITY 3: SAR chip (2×512×512)
SAR chip
2 × 512 × 512
CNN
(e.g., ResNet-18)
[64]
FUSION (per node)
concat
128 + 128 + 64 = 320
Linear
[320 → 192]
Linear + ReLU
[192 → 128]
[128]
One vector per node
MODALITY 4: RIVER GRAPH — 51 nodes
Flow edges (directed)
Upstream → downstream
(one direction)
Spatial edges (weighted)
Nearby nodes
weight = exp(−d / 40 km)
Flow edge (directed)
Spatial edge (weighted)
Node
Node features (one vector per node)
1 2 3 51
Graph Attention Network (GATv2)
• Flow edges (directed)
• Spatial edges (weighted)
• Multi-head attention
× 2 layers
Updated node embeddings
(one vector per node)
1 2 3 51
MLP (per node)
Linear → ReLU → Linear
P(flood in next 24h)
Calibrated probability
(per node)
1
0
Calibrated using
temperature scaling /
reliability calibration

4.3 Multimodal branch readiness

The SAR design is conceptually sound: fixed footprints enable dry-to-flood-to-recede sequences, and SAR can observe through monsoon cloud. The branch is not yet ready for network-wide claims. It currently has 321 frames across three Kelani sites, including incompatible 256 x 256 leftovers, while the expected completed branch is approximately 780 frames and the joined image_dataset.csv is not yet written.

The defensible use case is a KEL_HAN-focused case study or an ablation that treats SAR as an optional modality. The primary lead-time signal should still come from the hydro-meteorological branch because Sentinel-1 revisit is approximately 12 days; imagery is better positioned as current-state, morphology, and severity sharpening.

4.4 Evaluation and calibration feasibility

The evaluation design is stronger than a conventional accuracy-first setup. PR-AUC is the headline ranking metric because the primary positive rate is approximately 1.9%. Calibration must be reported using Brier score, ECE, reliability diagrams, and Brier decomposition, with temperature scaling or isotonic regression fit on the validation block only. Deep ensembles of five seeds are recommended for predictive variance.

The model should be compared against persistence, train-period climatology, a discharge-percentile rule, gradient-boosted trees, per-node LSTM/GRU, spatial-only GNN, and the full flow-plus-spatial model. This baseline ladder is necessary to distinguish genuine graph value from gains caused by strong discharge features or leakage.

4.5 Feasibility conclusion for the technical design

Technical finding: The core hydro-meteorological benchmark and graph model are feasible to implement from the current artifacts. The SAR branch is feasible as a controlled case study, but not yet as a network-wide multimodal claim. The first model milestone should therefore be tabular and graph baselines, followed by calibration, then SAR scalar and CNN ablations.

5. Evaluation protocol and decision gates

5.1 Required reporting set

Category	Mandatory outputs
Ranking	PR-AUC (headline), ROC-AUC
Calibration	Brier score, ECE with 15 bins, reliability diagram, Brier decomposition
Operational	POD/recall, FAR, CSI, F1 at validation-selected thresholds, precision-recall at fixed FAR budgets
Event level	Share of 1,469 events detected with at least one day of lead; mean lead time on detected events

Category	Mandatory outputs
Forbidden shortcut	Do not report plain accuracy as a primary result; a no-flood classifier would appear strong at this prevalence

5.2 Work packages and exit criteria

Work package	Scope	Exit criterion
WP1: Dataset freeze	Remove nine incompatible 256 x 256 frames; decide the 2024 core panel and 2025 SAR-only treatment; update stale docs and .gitignore.	A reproducible data freeze with current counts and no unresolved shape mismatch.
WP2: Baseline benchmark	Run persistence, climatology, discharge-percentile, tree model, and per-node sequence model under identical valid_sample filtering.	Baseline table with temporal, Gin, and event-level protocols.
WP3: Graph value	Compare no-graph, spatial-only, flow-only, and relational flow-plus-spatial variants.	Ablation showing whether topology adds value beyond node histories.
WP4: Calibration	Fit temperature or isotonic scaling on validation only; run ensemble or MC-dropout uncertainty.	Reliability and uncertainty package; no test-set calibration.
WP5: SAR case study	Complete v2 build or restrict to KEL_HAN; compare no imagery, SAR scalars, and SAR CNN.	An honest estimate of incremental SAR value with coverage mask reported.
WP6: External validity	Compare predictions with documented events and, when available, gauges or observed inundation.	Decision on research publication readiness and whether any operational pilot is defensible.

5.3 Acceptance language for the final paper or thesis

A defensible final claim is: the project constructs a reproducible, topology-aware benchmark and evaluates probabilistic flood prediction under leakage-controlled temporal, basin, and event protocols. A stronger claim that the model improves public flood warnings should be withheld unless gauge and observed-inundation evidence support it.

6. Risk register and limitations

6.1 Principal risks and controls

Risk	Why it matters	Control
Proxy labels	Discharge exceedance is not mapped inundation; urban drainage flooding can diverge.	Call labels proxy states; validate against inundation polygons and gauges before operational claims.
Reanalysis error	GloFAS and ERA5-Land can miss small, flashy, or reservoir-regulated catchments.	Report the three documented misses and stratify performance by basin and event type.
Leakage	Random node-day splits can leak correlated flood days across partitions.	Use precomputed temporal, Gin, and event protocols; use random splits only as a diagnostic inflation experiment.
Class imbalance	Primary positives are only 1.907% of valid rows.	Use focal or weighted losses; headline PR-AUC and event metrics, not accuracy.

Risk	Why it matters	Control
Quiet validation block	Validation positive rate is 0.810%, below train and test.	Use event-level CV and discuss model-selection bias.
Ragged 2025 edge	Only three nodes have 2025 coverage.	Truncate core experiments at 2024 or mask the edge explicitly.
SAR coverage	Only three Kelani sites have the v2 branch; build is incomplete.	Treat as case study; report presence masks and KEL_HAN-only results if needed.
No physical constraints	No conservation or hydraulic constraints are included in the loss.	Use terrain-aware / hydrology-grounded language, not physics-guided.
Data-service dependence	Public APIs and remote catalogs can change or rate-limit.	Cache raw data, pin code and metadata, and publish compact artifacts plus rebuild instructions.

7. Roadmap and final recommendation

7.1 Roadmap

Priority	Roadmap action
1	Finish the Branch B v2 fetch and publish image_dataset.csv.
2	Build Sentinel-1 change-detection flood masks for observed inundation cross-checks.
3	Add external flood polygons from World Bank, UNU, or UNESCO-IHP-WINS where licensing and coverage permit.
4	Integrate Sri Lanka Irrigation Department gauge and discharge records; this is the key upgrade for stronger physical validation.
5	Add 3-6 hour targets from hourly IMERG/GPM ingestion to move toward nowcasting.
6	Learn routing lags on flow edges to represent upstream-to-downstream travel time.

Appendix A. Node network summary

Basin	Nodes	Node IDs upstream to outlet
Kelani	7	KEL_UP, KEL_KIT, KEL_RUW, KEL_HAN, KEL_AVI, KEL_KAD, KEL_COL
Kalu	5	KAL_BAL, KAL_RAT, KAL_BUL, KAL_AGA, KAL_KLT
Mahaweli	7	MAH_NUW, MAH_PER, MAH_KAN, MAH_VIC, MAH_MHY, MAH_MAN, MAH_POL
Gin	4	GIN_NEL, GIN_THA, GIN_BAD, GIN_GAL
Nilwala	4	NIL_PIT, NIL_URA, NIL_AKU, NIL_MAT
AttanagaluOya	4	ATT_NIT, ATT_VEY, ATT_GAM, ATT_JAE
Batticaloa	3	BAT_VAL, BAT_ERA, BAT_BAT
DeduruOya	3	DED_KUR, DED_WAR, DED_CHI
Walawe	3	WAL_EMB, WAL_UDA, WAL_AMB
GalOya	2	GAL_ING, GAL_AMP
MahaOya	2	MHO_GIR, MHO_MOU
MalwathuOya	2	MAL_ANU, MAL_MAN
ColomboMetro	2	COL_KOT, COL_KOL
KirindiOya	1	KIR_TIS
KalaOya	1	KLA_ELA
YanOya	1	YAN_HOR

Appendix B. Current artifact status

Artifact area	Status
Core tabular dataset	Measured and described; 410,931 rows and 409,350 valid supervised samples.
Graph artifacts	nodes.csv and edges.csv specified; 35 flow and 204 spatial edges.
Validation report	11 documented flood episodes; misses retained and explained.
SAR v1 artifact	image_manifest.csv retained; fetched chips and image_index.csv superseded/deleted.
SAR v2 build	321 frames present; 9 incompatible 256 x 256 leftovers; image_dataset.csv not yet written.
Documentation	README.md and IMAGE_DATASET.md contain superseded claims and should be updated before release.
Git / distribution	Heavy rebuildable data excluded from git; release archive or Zenodo is the intended distribution path.

Appendix C. Source basis and references

This feasibility report is derived from the supplied Project Brief - Complete Handoff Pack, which records repository measurements as of 1 August 2026. The following references and data products are named in that brief and should be cited in any resulting paper or thesis:

- FloodGNN-GRU: a spatio-temporal graph neural network for flood prediction (Cambridge Environmental Data Science, 2024).
- Explicit Water Balance Constraints for Trustworthy Graph Neural Network Flood Forecasting (MDPI Applied Sciences, 2026).
- Roberts et al., Cross-validation strategies for data with temporal, spatial, hierarchical or phylogenetic structure (Ecography).
- Choosing blocks for spatial cross-validation: lessons from a marine remote sensing case study (Frontiers in Remote Sensing).
- Flood Uncertainty Estimation Using Deep Ensembles (MDPI Water, 2022).
- Real-Time Probabilistic Flood Forecasting Using Multiple Machine Learning Methods (MDPI Water, 2020).
- Intelligent flood forecasting and warning: a survey (OAE Intelligent Robotics / IEEE survey).
- Copernicus Emergency Management Service / GloFAS global river discharge reanalysis.
- ECMWF ERA5-Land reanalysis; NASA POWER daily point data; Microsoft Planetary Computer Sentinel-1 RTC; Copernicus DEM.